CURRICULUM VITAE

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EDUCATION:

1991-1995 **Cornell University** *Ithaca, NY.* B.S. in Biochemistry awarded May 1995.

1997-2004 **Columbia University College of Physicians & Surgeons** (1997-2004), *New York, NY. Integrated Program in Cellular, Molecular, and Biophysical Studies*,
Thesis Advisor: Jean Gautier
M.S. and M. Phil. awarded in January 1999
PhD awarded May 2004, *with distinction*.

POST-GRADUATE TRAINING:

2004-2009 **The Rockefeller University**, The Laboratory of Chromatin Biology, *New York, NY.*
Postdoctoral Fellow, Mentor: C. David Allis

PROFESSIONAL EMPLOYMENT:

2009-present **Albert Einstein College of Medicine**

2009-2015	*Assistant Professor*, Department of Biochemistry
2014-present	*Member*, Montefiore-Einstein Comprehensive Cancer Center
2015-2021	*Associate Professor,* Department of Biochemistry
2021-present	*Professor with Tenure*, Department of Biochemistry
2026-present	*Member,* Data Science Institute
2026-present	*Assistant Dean for Graduate Educational Innovation (as of 7/1/26)*

AWARDS AND HONORS:

2022	Julius Marmur Mentorship Award
2021	Irma T. Hirschl Career Development Award
2018	LaDonne H. Schulman Award for Excellence in Teaching
2018	Julius Marmur Mentorship Award
2013	Robbie Sue Mudd Kidney Cancer Research Scholar
2005	Irma T. Hirschl Foundation Fellowship
2005	NIH National Research Service Award
2004	PhD with Distinction, Columbia University

OTHER PROFESSIONAL ACTIVITIES:

Grant Review and Study Sections:

- *Ad hoc* Member, NIH ZRG1-CB-K55 Study Section, March 2020
- *Ad hoc* Member, NIH ZRG1 CB-I55 Study Section, March 2021
- *Ad hoc* Member, NSF BIO Directorate Review Panel, April 2026
- *Ad hoc* Member, NIH ZRG1 BCTB-J 57 R Special Emphasis Section, July 2026
- *2030 Einstein Acceleration Fund* Review Panel, ad hoc and discussion panel reviewer, May 2026
- International Grant Reviewer for DFG (Germany), Wellcome Trust/India, ANR (France), Wellcome Trust (UK), UKRI, US-Israel National Science Foundation Binational Grant, and local Grant Reviewer for City University of New York, NSF and NIH mail reviewers for individual grants

Regional:

- Lead organizer of a Scientific Symposium at The Rockefeller University, NY, Oct 2011, "Deciphering the Allis Code"
- Founding organizer of new NYC Chromatin Club (www.nycchromatinclub.org, built @nycchromatin twitter and slack channel to develop community) (2020-present)
- Lead Organizer and Fundraiser, July 2025 NYC Chromatin Club meeting at Albert Einstein College of Medicine (raised $21,000 in corporate sponsorship)

Editor:

- Issue of *Cellular and Molecular Life Sciences* (Springer, Basel, Switzerland, IF ~ 5.8) focused on "Protein Methylation in Cellular Physiology" for 2019. I solicited and edited 7 review manuscripts. Published Summer 2019: https://link.springer.com/journal/18/76/15

Ad Hoc Reviewer:

- Nature, NSMB, Genes & Development, Cell Reports, Molecular Cell, Developmental Cell, Journal of Protein Science, Journal of the American Cancer Society, Biotechniques, BBA, Journal of Cell Science, ACS Med Chem, Developmental Cell, Cell Reports, JACS, Nucleic Acids Research, PLoS ONE, many others (*average 5 journal peer reviews/year*)

Professional Consulting:

- Scientific Advisory Board, *Epinova Therapeutics, Inc.* (2014-2015)
- Consultant for *SK Biopharmaceuticals* (2017, 2018)
- Consultant for *Prelude Therapeutics* (2020)

Teaching:

- Course Leader/Director, *Gene Expression: Beyond the Double Helix PhD/MSTP course* (5-credit core annual Block II graduate school course), 2012-2026
- Lecturer, Discussion Group facilitator, *Gene Expression: Beyond the Double Helix PhD/MSTP course, 2011-present,* annual
- Lecturer, *Biochemistry* (3 or more lectures), 2009-present, annual

- Lecturer, Discussion Group facilitator, *Chemical Biology,* 2023-present, annual

Department and College Service:

- Co-founder (2011) and Lead Organizer (2011-2015) for **"Einstein Chromatin Club"** Joint Lab Meetings at Einstein, with the Gamble, Secombe, Coleman, Skoultchi, Sidoli, Stengel and other labs, biweekly meetings (2011-present)
- Biochemistry Representative, Graduate Executive Committee, 2018-present
- Member, Associate Professor Promotion Committee, 2016, 2017
- Member, PhD Admissions Committee, 2016
- Member, Science Council, Communications and Public Affairs, 2017-2022
- Co-Founder and steering committee member, Einstein Chromatin Club (2011-present)
- Chair, Biochemistry Department External Seminar Series, 2016-present
- Chair, Biochemistry Department Student Seminar Series, 2016-2023
- Member, Biochemistry Department faculty search committees, 2016, 2017, 2021, 2023, 2026
- Member, Middle States Commission on Higher Education, Accreditation Self-Study, Working Group III, 2019
- Biochemistry Faculty Representative, Departmental Reorganization Task Force, 2020
- Member, Dean's Reorganization and Research working group, with McKinsey, 2020
- Member, Albert Einstein Cancer Center Faculty Recruitment *ad hoc* Committee on Cancer Epigenetics, 2020-2022
- Member, Graduate Executive Committee (2018-present)
- Co-Chair, Graduate Curriculum Committee (2022-2023)
- **Chair, Graduate Curriculum Committee (2023-present)**
- Member, Discovery Working Group, Einstein Strategic Planning Committee (2024)
- Member, Einstein Faculty Awards Committee (2024- 2025)
- Chair, Graduate Curriculum Committee – Sub-Committee on responsible use of Generative Artificial Intelligence (2024)
- Member, Committee on Ethical Research Performance (2024-present)
- Member, Strategic Planning Committee Subcommittee on Faculty Compensation for Teaching (2024-present)
- Member, Professor Promotion Committee (2024, 2025)
- Co-Chair, Dean's Committee on Artificial Intelligence Policy and Guidance (2025)
- Co-Chair, Strategic Plan Implementation Education Pillar, Artificial Intelligence in Education Initiative (2025-present)
- Co-Chair, Standing Artificial Intelligence Review Committee (ARC) (2025-present)
- Co-Chair, Working Group for Middle States for Standard 5: Educational Effectiveness Assessment (2025-present)
- Member, Montefiore Einstein Comprehensive Cancer Center (MECCC) Strategic Plan Working Group on Education and Training; Leading Data Science Institute initiative for training (2026-present)

Community Outreach and Service

- Youth basketball coach, 5th and 6th grades, WJBL 2015, 2016
- Speaker at American Cancer Society Relay for Life, Harrison, NY 2017
- Mt. Vernon Boys and Girls Club – career path speaker, March 2019
- Beth El Synagogue Center, New Rochelle Trustee (2016-present), Vice President (2019-2023), Executive Vice President (2023-present), President (starting July 2026)

RESEARCH:

Current Grant Support

Irma T. Hirschl Trust Career Award
"*Protein arginine methyltransferases in cancer"*
05/2021-12/2025
Direct costs: $40,000/yr

Venture Capital Pitch Prize
"Development of a novel enzyme inhibitor"
09/2022-no end date
Role: PI
Direct Costs: $25,000

ALS Therapy Development Institute
"Role of Protein Arginine Methyltransferases in ALS-mutant C9orf72 Cellular Cytotoxicity"
01/2024-12/2026
Total Costs: $435,734

Hevolution Foundation
"GNMT as a novel therapeutic target for aging"
05/2024-04/2029
Role: Co-PI (PI: Derek Huffman, Co-PI Evripidis Gavathiotis)
Direct costs: $500,000/yr

Einstein 2030 Acceleration Fund
"Targeting the TTLL4 glutamyltransferase as a first-in-class NPM1c-mutant AML therapeutic"
06/2025-05/2026
Role: PI (Co-I, Kira Gritsman, Seiya Kitamura)
Direct Costs: $119,500

MECCC Transdisciplinary Science Grant
"Targeting the TTLL4 glutamyltransferase as a first-in-class NPM1c-mutant AML therapeutic"
06/2025-05/2026
Role: PI (Co-I, Seiya Kitamura)
Direct Costs: $100,000

NIH/NIA R61AG096065
"Connecting surgical stress to sex-dependent vulnerabilities in aging and aging hallmarks"
09/2025-08/2027
Role: Collaborator (PI: Derek Huffman)
5% effort

Completed Grant Support

CA200503 - Department of Defense Peer Reviewed Cancer Program Idea Award
"NPM1c Post-Translational Glutamylation in Leukemogenesis"
Role: PI, Percent Effort 10%, (Co-I Kira Gritsman, MD, PhD, *Cell Biology* and *Medicine*)
09/2021-08/2024
Direct costs: $166,667/yr

R01GM108646 NIH/NIGMS
"Crosstalk to Histone Arginine Methylation"
9/2014-8/2024; Competitively renewed in 2019
Role: PI, Percent Effort: 30%
Direct costs: $206,000/yr

R01GM135614 NIH/NIGMS
"Regulation of Histone Chaperone Function by Glutamylation"
1/2020-12/2024
Role: PI, Percent Effort: 25%
Direct Costs: $220,000/yr

R01GM108646-09S1 NIGMS/NIH
Supplement funding the purchase of a new Keyence BZ-X810 fluorescence microscope
09/2021-09/2022
Role: PI
Direct Costs: $165,175

Sponsored Research – Cambrian Biopharma
Screening of a methyltransferase for activators
Role: Co-I, with Derek Huffman and Evris Gavathiotis
10/2020-09/2021
Direct costs: $152,348
5% effort (Shechter), 25% effort for a lab member for 1 year

American Cancer Society Research Scholar Award
"Regulation of Histone Chaperone Function by Post-Translational Modification"
01/2014-12/2017
Role: PI
Direct costs: $146,000/yr

Alexandrine and Alexander Sinsheimer Foundation Research Scholar
"Exploring the Role of Histone Arginine Methylation in Breast Cancer"
08/2011-07/2014
Role: PI.
Direct costs, $50,000/yr

HHS-0007-16SF SAS Foundation
"*Targeting the PRMT5 Arginine Methyltransferase: An Achilles' Heel for Many Cancers*"
01/2018-12/2018
Role: PI
Direct Costs: $50,000

R01GM108646-05S1 NIGMS/NIH
Supplement funding the purchase of a new SPR instrument
09/2018-08/2019
Role: PI
Direct Costs: $84,696

American Lung Association Discovery Award

“*When the Writing Goes Wrong: Protein Arginine Methyltransferases in Lung Cancer”*
07/2018-12/2020
Role: PI, Percent Effort, 5%
Direct Costs: $100,000/yr

Albert Einstein Cancer Center Pilot Award
“NPM1c glutamylation in AML”
01/2020-12/2021
Role: Co-PI, with Kira Gritsman
Direct Costs: $23,683

R01GM135614-051S1 NIGMS/NIH
Supplement to purchase a new chromatography system
Role: PI
07/2020-06/2021
Direct costs: $96,769

R01GM108646-03S1 NIGMS/NIH
Purchase of chromatography supplies
09/2016-08/2017
Direct Costs: $14,103

Irma T. Hirschl Postdoctoral Fellowship
2005-2006
Role: PI (Postdoc)

F32GM075486-01, NIGMS/NIH
“Epigenetic Mechanisms: Inheritance of Histone States”
03/2006-02/2009
Role: PI (Postdoc)

Funding awarded to mentored students

Trainer, Cellular and Molecular Biology Training grant
Christopher Warren, 2014-2016 (reappointed 3 years)

Trainer, Medical Scientist Training Program Grant
Maxim Maron, 2017-2019 (reappointed 3 years)
Joseph DeAngelo, 2020
Jacob Roth, 2021
Isaac Kraz, 2023
Liana Valin, 2026

NIH/NIGMS Predoctoral Fellowship F31GM116536
Christopher Warren, 2016-2018 (2 years)

INVITED NON-EINSTEIN PRESENTATIONS (Since 2010):

- Feb **2010**. “Writing the Embryonic Histone Code”. Mt Sinai School of Medicine.
- October **2011**. “Defining the Nascent Transcriptome During Zygotic Gene Activation in Xenopus

laevis", Woods Hole Oceanographic Institute, Xenopus National Resource Development Meeting
- October **2012**. "Writing the Embryonic Histone Code", SUNY New Paltz.
- October **2013**. "Writing the Embryonic Histone Code", Microbiology and Mol Genetics, University of Pittsburgh.
- May **2013**. "Novel Mechanism of Substrate Recognition by the Arginine Methyltransferase PRMT5-MEP50", GTC Epigenetics in Drug Discovery, Boston, MA; *Chair, keynote session*
- August **2015**. "Regulation of Histone Deposition by Post-translational modification". Gordon Research Conference on Chromosome Dynamics.
- November **2016**. Miami Epigenetics and Cancer Symposium (Miami, FL)
- November **2016**. UT Southwestern, Dallas, Texas – "How to Write a Histone Code"
- October **2017** Mt. Sinai School of Medicine, NY Department of Oncological Sciences.
- October **2017** Academia Sinica Institute of Biological Chemistry, Taipei, Taiwan
- November **2017**. University of North Carolina, Chapel Hill.
- August **2018**, Mt. Sinai, Structural Biology Discussion Group, NY.
- March **2019**, Oxford University, Weatherall Institute of Molecular Medicine, Oxford, UK
- March **2019**, University of Nottingham, England, UK
- September **2019**, University of Michigan Pathology Department Seminar Series
- *Postponed due to COVID*: **originally scheduled for October 2020,** Nanjing Medical University, Nanjing, China (host: Hongshan Chen)
- *Postponed due to COVID*, Ben Gurion University, Beersheva, Israel (host: Danny Levy)
- *Postponed due to COVID***,** Tsinghua School of Medicine, Beijing, China (host: Haitao Li)
- May **2021**, Abcam Meeting on Protein Arginine Methylation: Recent advances in basic and translational studies, invited speaker
- November **2021**, Epicypher Meeting, Clearwater Florida, invited speaker
- February **2023,** FASEB Biological roles of Arginine Methylation, invited speaker (online)
- November **2023**, Epicypher Meeting, invited speaker
- February **2024**, FASEB Biological roles of Arginine Methylation, invited speaker (online)
- March **2024**, University of Toronto, invited speaker
- July **2024**, FASEB Meeting on Biological Methylation, Porto, Portugal (oral presentation)
- March **2025**, MD Anderson Comprehensive Cancer Center, invited speaker
- May **2025**, Syosset High School Science Research Program Invited Keynote speaker
- November **2025**, Epicypher Meeting on Epigenetics & Chromatin, invited speaker
- December **2025**, Data Science Institute AI Series, Einstein, invited speaker
- January **2026,** FASEB Meeting on Arginine Methylation (Japan), invited speaker
- March **2026**, St. John's University, invited speaker
- June **2026**, Albert Einstein College of Med, Department of Medicine Grand Rounds, invited speaker
- June **2026,** FASEB Biological Methylation (Ireland), invited speaker
- Oct **2026**, ALS-TDI Research Summit (Boston), invited speaker (*forthcoming*)
- July **2027,** FASEB Meeting on Arginine Methylation (Korea), invited speaker (*forthcoming)*

RESEARCH HIGHLIGHTS 2024-2026

- Identified GRIPPs (Genomically Retained Incompletely Spliced Transcripts) as critical intermediates in regulating RNA stability and nuclear retention, revealing a new layer of gene expression control and relevant to cancer therapy with PRMT5 inhibitors.

- Defined PRMT5-dependent arginine methylation events controlling splicing factor interactions and RNA processing, advancing our mechanistic understanding of epigenetic regulation in gene expression.

- Discovered that glutamylation of NPM1c modulates its oncogenic activity in AML by influencing chromatin accessibility and transcriptional programs, highlighting a novel therapeutic target in leukemia.

BIBLIOGRAPHY

A. Original Communications in Peer-Reviewed Journals (Chronological Order):

1. Yao, N., Turner, J., Kelman, Z., Stukenberg, P., Dean, F., **Shechter, D.**, Pan Z., Hurwitz, J., and O'Donnell, M. **1996**. Clamp loading, unloading and intrinsic stability of the PCNA, Beta, and gp45 sliding clamps of human, E. coli, and T4 replicases. *Genes to Cells*. 1:101-113.

2. **David Shechter,** C.Y. Ying, and J. Gautier. **2000**. The intrinsic DNA helicase activity of *Methanobacterium thermoautotrophicum* delta H minichromosome maintenance protein. *J.Biol. Chem.,* 19;275:15049-15059.

3. Vincenzo Costanzo, **David Shechter**, Patrick J.Lupardus, Karlene A.Cimprich, Max Gottesman, and Jean Gautier. **2003**. An ATR-and Cdc7-dependent DNA damage checkpoint that inhibits initiation of DNA replication. *Molecular Cell.* 11:203-213.

4. **David Shechter**, Vincenzo Costanzo, and Jean Gautier. **2004**. ATR and ATM Regulate the Timing of DNA Replication Origin Firing. *Nature Cell Biology.* 6, 648-655.*
    a. **accompanied by a News and Views article*

5. **David Shechter**, Carol Ying, and Jean Gautier. **2004**. DNA Unwinding is an MCM-Complex and ATP-Hydrolysis Dependent Process. *J Biol Chem*, 44;279:45586-45593.

6. **David Shechter*,*** Holger L. Dormann, C. David Allis, and Sandra B. Hake. **2007**. Extraction, Purification, and Analysis of Histones. *Nature Protocols*. 2007. 2, 1445-1457.

7. **David Shechter*,*** Raghu Chitta, Joshua J. Nicklay, Jeffrey Shabanowitz, Donald F. Hunt, and C. David Allis. **2009.** Analysis of histones in *Xenopus laevis* Part I: A distinct index of enriched variants and modifications exists in each cell type and is remodeled during developmental transitions**.** *J Bio. Chem,* 284:1064-74.

8. Joshua J. Nicklay*, **David Shechter****,* Raghu Chitta, Ben Garcia, Jeffrey Shabanowitz, Donald F. Hunt, and C. David Allis. **2009.** Analysis of histones in *Xenopus laevis* Part II: Mass Spectrometry reveals an Index of Cell-type Specific Modifications on H3 and H4. *J Biol Chem, 284:1075-85.*
    a. *co-1st author

9. Andrew Xiao, Haitao Li, **David Shechter**, Sung Hee Ahn, Laura A. Fabrizio, Hediye Erdjument-Bromage, Satoko Murakami-Ishibe, Bin Wang, Paul Tempst, Kay Hofmann, Dinshaw J. Patel, Stephen J. Elledge and C. David Allis. **2009.** WSTF regulates the DNA damage response of H2A.X via a novel tyrosine kinase activity. *Nature,* 457:57-62.

10. **David Shechter**, Raghu K. Chitta, Andrew Xiao, Jeffrey Shabanowitz, Donald F. Hunt, and C. David Allis. **2009**. A Distinct H2A.X Isoform is Enriched in *Xenopus laevis* Eggs and Early Embryos and is Phosphorylated in the Absence of a Checkpoint. *Proc Natl Acad Sci,* 106:749-54.

11. Laura Banaszynksi, C. David Allis, and **David Shechter**, **2010**. "Analysis of Histones and Chromatin in *Xenopus laevis* Egg and Oocyte Extracts." *Methods.* 51:3-10*.*

12. Carola Wilczek, Raghu Chitta, Eileen Woo, Jeffrey Shabanowitz, Brian Chait, Donald Hunt, and **David Shechter. 2011**. "Protein arginine methyltransferase PRMT5/MEP50 methylates histones H2A and H4 and the histone chaperone Nucleoplasmin in Xenopus laevis eggs." *Journal of Biological Chemistry,* 286:42221-31.

13. Meng-Chiao Ho, Carola Wilczek, Jeffrey Bonnano, Li Xing, Janina Seznec, Tsutomu Matsui, Lester G. Carter, Takashi Onikubo, P. Rajesh Kumar, Michael Brenowitz, R. Holland Cheng, Ulf Reimer, Steven Almo, **David Shechter**. **2013.** "Structure of the Arginine Methyltransferase PRMT5-MEP50 Reveals a Mechanism for Substrate Specificity". *PLoS ONE 8(2): e57008. doi:10.1371/journal.pone.0057008*

14. Wei-Lin Wang, Lissa C. Anderson, Joshua J. Nicklay, Hongshan Chen, Matthew J. Gamble, Jeffrey Shabanowitz, Donald F. Hunt, **David Shechter 2014.** "Phosphorylation and arginine methylation mark histone H2A prior to deposition during Xenopus laevis development". *Epigenetics and Chromatin,* 7:22. doi:10.1186/1756-8935-7-22.

15. Emmanuel S. Burgos, Carola Wilczek, Takashi Onikubo, Jeffrey B. Bonanno, Janina Jansong, Ulf Reimer, and **David Shechter**. **2015**. "Histone H2A and H4 N-Terminal Tails are Positioned by the MEP50 WD-Repeat Protein for Efficient Methylation by the PRMT5 Arginine Methyltransferase." *Journal of Biological Chemistry*. PMID: 25713080 doi: 10.1074/jbc.M115.636894

16. Takashi Onikubo, Joshua J. Nicklay, Li Xing, Christopher Warren, Brandon Anson, Wei-Lin Wang, Emmanuel S. Burgos, Sophie E. Ruff, Jeffrey Shabanowitz, R. Holland Cheng, Donald F. Hunt, and **David Shechter**. **2015.** "Developmentally Regulated Post-Translational Modification of Nucleoplasmin Controls Histone Sequestration and Deposition". *Cell Reports*, Mar 11. doi:10.1016/j.celrep.2015.02.038, PMID: 25772360

17. Sun J, Zhao Y, McGreal R, Cohen-Tayar Y, Rockowitz S, Wilczek C, Ashery-Padan R, **Shechter D,** Zheng D, Cvekl A. "Pax6 associates with H3K4-specific histone methyltransferases Mll1, Mll2, and Set1a and regulates H3K4 methylation at promoters and enhancers." *Epigenetics and Chromatin.* **2016** Sep 9;9(1):37. doi: 10.1186/s13072-016-0087-z. PMID: 27617035 PMCID: PMC5018195

18. Wei-lin Wang and **David Shechter. 2016**. "Chromatin assembly and transcriptional cross-talk in *Xenopus laevis* oocyte and egg extracts." *Int J. Dev. Biol*. doi: 10.1387/ijdb.160161ds

19. Hongshan Chen, Benjamin Lorton, Varun Gupta, **David Shechter. 2017**. "A TGFβ-PRMT5-MEP50 axis regulates cancer cell invasion through histone H3 and H4 arginine methylation coupled transcriptional activation and repression." *Oncogene*, 36, p373–386 doi: 10.1038/onc.2016.205, PMID: 27270440 PMCID: PMC5140780

20. Emmanuel S. Burgos, Ryan O. Walters, Derek M. Huffman, and **David Shechter. 2017**. "A simplified characterization of S-adenosyl-L-methionine-consuming enzymes with 1-Step EZ-MTase: a universal and straightforward coupled-assay for in vitro and in vivo setting." *Chem. Sci*. doi: 10.1039/C7SC02830J

21. Christopher Warren, Tsutomu Matsui, Jerome M. Karp, Takashi Onikubo, Sean Cahill, David Cowburn, Michael Brenowitz, Mark Girvin and **David Shechter**, "Dynamic Intramolecular Regulation of the Histone Chaperone Nucleoplasmin Controls Histone Binding and Release".

Nature Communications, **2017**;8(1):2215. doi: 10.1038/s41467-017-02308-3. PMID: 29263320 PMCID: PMC5738438

22. Wei-Lin Wang, Takashi Onikubo, and **David Shechter**. **2018**. "Chromatin Characterization in *Xenopus laevis* cell free egg extracts and embryos". *CSHL Protocols. doi:* 10.1101/pdb.prot099879

23. Ryan O. Walters, Esperanza Arias, Antonio Diaz, Emmanuel S. Burgos, Fangxia Guan, Simoni Tiano, Kai Mao, Cara L. Green, Yungping Qiu, Hardik Shah, Donghai Wang, Adam Hudgins, Tahmineh Tabrizian, Valeria Tosti, **David Shechter**, Luigi Fontana, Irwin J. Kurland, Nir Barzilai, Ana Maria Cuervo, Daniel E.L. Promislow, and Derek M. Huffman. **2018**. "Sarcosine is uniquely modulated by aging and dietary restriction in rodents and humans". *Cell Reports.* 2018 Oct 16;25(3):663-676.e6. doi: 10.1016/j.celrep.2018.09.065.

24. Mirunalini Ravichandran, Run Lei, Qin Tang, Yilin Zhao, Joun Lee, Liyang Ma, Stephanie Chrysanthou, Benjamin M.Lorton, Ales Cvekl, **David Shechter**, Deyou Zheng, Meelad M. Dawlaty. "Rinf Regulates Pluripotency Network Genes and Tet Enzymes in Embryonic Stem Cells". *Cell Reports* Volume 28, Issue 8, 20 August **2019**, Pages 1993-2003.e5, doi:10.1016/j.celrep.2019.07.080

25. Benjamin M. Lorton, Rajesh K. Harijan, Emmanuel S. Burgos, Jeffrey B. Bonanno, Steven C. Almo, and **David Shechter.** "A binary arginine methylation switch on histone H3 Arginine 2 regulates its interaction with WDR5." *Biochemistry*, DOI: 10.1021/acs.biochem.0c00035 • Publication Date (Web): 24 Mar **2020**

26. Christopher Warren, Jeffery B. Bonanno, Steven C. Almo, **David Shechter 2020**. "Structure of a Single Chain H2A/H2B Dimer". *Acta Cryst.* (2020). F76, 194-198, doi: 10.1107/S2053230X20004604

27. Maron MI, Lehman SM, Gayatri S, DeAngelo JD, Hegde S, Lorton BM, Sun Y, Bai DL, Sidoli S, Gupta V, Marunde MR, Bone JR, Sun ZW, Bedford MT, Shabanowitz J, Chen H, Hunt DF, **Shechter D**. Independent transcriptomic and proteomic regulation by type I and II protein arginine methyltransferases. *iScience***. 2021** 11;24(9):102971. doi: 10.1016/j.isci.2021.102971. PMCID: PMC8417332.

28. Maxim I Maron, Alyssa D Casill, Varun Gupta, Jacob S Roth, Simone Sidoli, Charles C Query, Matthew J Gamble, **David Shechter.** "Type I and II PRMTs inversely regulate post-transcriptional intron detention through Sm and CHTOP methylation". *eLife*, **2022**; 11:e72867 DOI: 10.7554/eLife.72867, PMCID: PMC8765754

29. Benjamin M. Lorton, Christopher Warren, Humaira Ilyas, Prithviraj Nandigrami, Subray Hegde, Sean Cahill, Stephanie M Lehman, Jeffrey Shabanowitz, Donald F. Hunt, Andras Fiser, David Cowburn, **David Shechter**. "Glutamylation of Npm2 and Nap1 acidic disordered regions increases DNA mimicry and histone chaperone efficiency". *iScience* **2024**; doi:10.1016/j.isci.2024.109458

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PMCID: PMC12041856.

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32. Abin Biswas, Omar Muñoz, Kyoohyun Kim, Carsten Hoege, Benjamin M. Lorton, **David Shechter**, Jochen Guck, Vasily Zaburdaev, Simone Reber. "Conserved nucleocytoplasmic density homeostasis drives cellular organization across eukaryotes". *Nature Comms*, **2025.** 16, 7597 DOI: 10.1038/s41467-025-62605-0

33. Joseph D. DeAngelo, Maxim I. Maron, Jacob S. Roth, Subray Hegde, Aliza M. Silverstein, Varun Gupta, Stephanie Stransky, Joel Basken, Joey Azofeifa, Charles C. Query, Simone Sidoli, Matthew J. Gamble, **David Shechter**. "Productive mRNA Chromatin Escape is Promoted by PRMT5 Activity" *Mol Cell*, **2025**, Volume 85, Issue 21 p4016-4031.e9 PMID: 41086806

Preprints (unpublished):

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B. Review Articles, Book Chapters, and Other Contributions:

Peer-Reviewed Reviews and Commentaries:

39. **David Shechter** and Jean Gautier. **2004**. MCM Proteins and Checkpoint Kinases Get Together at the Fork. *PNAS*, 101, 10845-10846.

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**accompanied by cover image*

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49. Carola Wilczek, Janina Jansong, Ulf Reimer, **David Shechter**. "Use of High-Density Histone Peptide Arrays for Parsing the Specificity of a Histone-Modifying Enzyme Complex". JPT Innovative Peptide Solutions, 2013.

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Citation metrics:

Google Scholar Citations: 6068
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Intellectual property:

“Chemoenzymatic Synthesis of a SAM-Cofactor/SAH Library for Proficient Profiling of Protein Methyltransferases”. *U.S. Provisional Patent Application No. 62/290,502. Application no longer pending.*

“Characterization of S-adenosyl-l-methionine consuming enzymes with 1-Step EZ-MTASE: a Universal Coupled Assay”. *US Patent US20230220443A1*

“TARGETING AN ENZYME REQUIRED FOR ACUTE MYELOID LEUKEMIA*” Internal Einstein Disclosure AET-03260, U.S. PCT/US2023/024566*

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2. Wei-Lin Wang, PhD (postdoc , 2011-2013), currently Senior Manager at CureVac SE
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4. Christopher Warren, PhD (student, 2013-2018)*, currently Associate Principal Scientist at Merck
5. Benjamin Lorton (PhD student, 2015-2021* and postdoc 2021-2023), currently Drug Design Consultant
6. Hongshan Chen, PhD (Postdoc/Associate , 2016-2017), currently Professor Nanjing Medical University
7. Maxim Maron (MD/PhD student in my lab, 2017-2022)*, currently PSTP fellow MSKCC
8. Joseph DeAngelo (MD/PhD student, 2020-2025)*, currently PSTP resident, MGH
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10. Isaac Krasnopolsky (MD/PhD student, 2021-present)
11. Humaira Ilyas (postdoc, 2021-2025)
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16. Subray Hegde, PhD (Instructor, 2020- present)

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18. Sara Rouhanifard (2010)
19. Jason Ford (2010)
20. Penelope Ruiz (2012)
21. Yizhou Zhu (2012)
22. Micheal Lesser (2012)
23. Sean Healton (2013)
24. Chayim Goldberg (2013)
25. Karishma Smart (2015)
26. Prasoon Jaya (2018)
27. Harmony Ketchum (2019)
28. Victor Paulino (2019)
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31. Noah Jacobs (MSTP Rotation 2025)
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39. Ahava Muskat (SURP 2015)
40. Anniya Gu (SURP 2016)
41. Adam Haimowitz (SURP 2017)
42. Rachel Fortinksy (High School 2017)
43. Neda Shokrian (SURP 2018)
44. Philip Nagler (SURP 2019)
45. Caradonna, Jonathan (SURP 2022)
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54. Andrew Johnston (2015-2018, SAC and defense)
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59. Denis Ruiz (2018, thesis defense)
60. Nadege Gitego (2018-2023, SAC chair, thesis defense)

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68. Andrea Lopez (2020, thesis defense)
69. Adam Spitz (2020, thesis defense)
70. Miriam Ben-Dayan (2020, thesis defense)
71. Helen Jung (2021-2025, SAC, thesis defense)
72. Gabriel Bedard (2021-2024, SAC, thesis defense)
73. Mu Feng (2022, thesis defense)
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